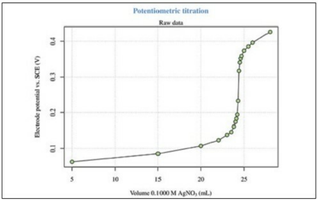

Figure 1: Potentiometric titration

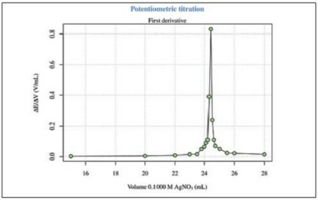

Figure 2: First derivative

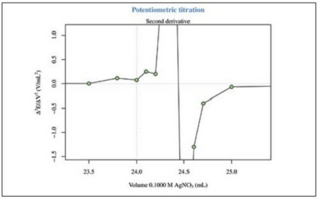

Figure 3: Second derivative with zoom

```
grid(...)
points(x2,y2,type="o")
```

Now, with a *zm(type="session")*, I can zoom on the plot (Figure 3) for a better visualisation and then, with *locator()*, I can obtain the inflection point coordinates.

Conductometric titration

For a general explanation of this technique, log on to *http://en.wikipedia.org/wiki/Conductometry*. The first experiment (Figure 4) is taken from *Reference 5* given at the end of the article and is about the conductometric titration curve for NaOH with 0.1219 M potassium hydrogen phthalate ($C_8H_5KO_4$). The second experiment (Figure 5) is also taken from the same reference, and is about the conductometric titration curve for H_3PO_4 with 0.5016 M NaOH. These two experiments are problems of linear segmented regression (see *Reference 6*) about the determination of the break-point(s). The most important aspect relates to the *psi* parameter within the *segmented* package call. The *psi* parameter includes starting values for the break-point(s) for the corresponding variable in *seg.Z*. If *seg.Z* includes only a variable, *psi* may be a numeric vector. With *summary(fit)* I can visualise the break-point(s) coordinates on the *x* axis. The easiest way to plot the regression lines is the use of *abline(intercept,slope,col=your-color)*, but the *abline* line type spans the whole plot width. The figures presented here have been arrived at with some more calculations and the regression lines are plotted with *lines*. The following code can be used to calculate the slope and the intercept for each segment:

```
fit<-segmented(lm(y~x),seg.Z=~x,psi=list(x=c(5)))   # Figure 4
a1=slope(fit)$x[1]
b1=intercept(fit)$x[1]
...
fit<-segmented(lm(y~x),seg.Z=~x,psi=list(x=c(3,6))) # Figure 5
a1=slope(fit)$x[1]
b1=intercept(fit)$x[1]
...
```

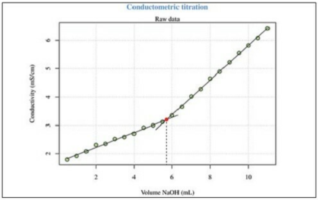

Figure 4: Conductometric titration (1)

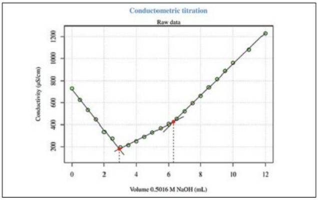

Figure 5: Conductometric titration (2)